

Conservation of Shadows

Haru HyunKyung Ji
hji@ocadu.ca
OCAD University
Toronto, Canada

Graham D. Wakefield
grrrwaaa@yorku.ca
York University
Toronto, Canada



Figure 1: Artificial Nature: Conservation of Shadows in the ZKM. BioMedien: The Age of Media with Life-like Behavior. ZKM Centre for Art and Media, Karlsruhe, Germany. 2021-2022. Photograph by Jonas Zilius.

ACM Reference Format:

Haru HyunKyung Ji and Graham D. Wakefield. 2023. Conservation of Shadows. In *Proceedings of SIGGRAPH '23 Art Gallery*. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588428.3593826>

1 PROJECT URL

<https://artificialnature.net>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Art Gallery, August 06-10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0157-3/23/08.

<https://doi.org/10.1145/3588428.3593826>

2 DESCRIPTION OF ARTWORK

Time is instantaneously crystallized in space to form glistening trails of you and other creatures within this interactive mixed reality installation, with rustling sounds and mingling shadows as shared physical connections between the visible and invisible realms of a bio-inspired computational ecosystem.

Conservation of Shadows is a mixed reality habitat where humans encounter unfamiliar life-like creatures in physical and virtual realms. Invisible and unknown beings seem to explore the space around you - ghostly shadows passing across the floor, chasing and mingling with your own shadows, and whistling past the small bells, making them ring. Entering virtual reality, you dive deeper into an artificial ecosystem cohabiting in superposition with your own. What first appeared as shadows on the floor are unveiled

to be an entire biome moving around us: snake-like entities behaving curiously, searching for anything new in the world, and smaller builder-creatures seeking out ephemeral traces of being and spinning threads as they go. These threads are in endless motion, weaving into longer spatial memories as a constantly evolving network conducting color and motion, like the neurons in a brain or the hyphae of a mycelial network. Neither completely programmed nor entirely random, they leave enough room for collectively and adaptively organized growth.

Here, sounds and shadows work as a bridge between physical and invisible realms. In this sensitively endogenous world, all sensorial components have intrinsic functions. All continuously build inter-relationships among themselves, including humans. Its defamiliarized nature awakens us to the weirder and wilder nature beyond the human-centered world-as-we-know-it; as a vitally playful and curious art-as-it-could-be in the age of the Anthropocene.

The computational biotope of Conservation of Shadows raises questions about how we may coexist with non-human beings in ways that are more abundantly curious and mutually rewarding. Each installation has drawn inspiration from the histories and hauntings of the exhibition venue – such as the volatile biomaterials of a former center for disease control in Korea, or the munitions manufacturing of a factory in Germany – embodied in the works as site-specific aspects of both physical and virtual materiality. These are capsules of time passed understood as seeds for the imaginative rethinking of the past and future, or rather, the superpositions of shorter or longer presents as loops of memory that evolve and rejoin in new forms. Similarly, the creatures’ weaving of space is driven through curiosity-based artificial intelligence not only as an effective means of discovery, but also because curiosity is the beginning of empathy: how I and you become we, and how we may envision worlds-that-could-be.

Technically, as an interactive and immersive mixed reality installation artwork, Conservation of Shadows involves a network of over fifty IoT motor-actuated sound objects, ceiling to floor projection, real-time volume capture, and a simultaneous motion tracked virtual reality perspective inhabited by artificial life. The majority of the software was custom designed by the artists, using a combination of C++, Javascript, 3D modeling and texturing, OpenGL and custom GLSL shaders. The artists’ software comprises all simulation of the world, including 3D volume field processes, creature kinetics and artificial intelligence, calibration and volume capture, networking, and graphical rendering for projection and virtual reality.

3 ARTIST INFORMATION

Haru Ji is an artist and founder of the art research project “Artificial Nature”. It is an art of artificial ecosystems as shared realities to shatter the perspective of humans as the center of the world and deepen our understanding of the complex intertwined connections in dynamic living worlds. Since 2007 Artificial Nature artworks have been installed at international venues including SIGGRAPH, ISEA, La Gaite Lyrique (Paris), ZKM (Karlsruhe), Microwave (Hong Kong), Currents (Santa Fe), and awarded in the VIDA Art & Artificial Life competition. Dr. Ji is an Associate Professor at OCAD University in Toronto, Canada.



Figure 2: Artificial Nature: Conservation of Shadows in the ZKM. BioMedien: The Age of Media with Life-like Behavior. ZKM Centre for Art and Media, Karlsruhe, Germany. 2021-2022. Photograph by Jonas Zilius.

Graham Wakefield’s research is dedicated to computational art practice and software development for computer music and mixed / hybrid reality. It applies a deep commitment to the open-endedness of computation—as an art material—as expressed both in live software for artists and musicians (such as the gen environment for Max/MSP), as well as immersive artworks of biologically-inspired systems (working with Haru Ji as Artificial Nature). He is currently an Associate Professor and Canada Research Chair in the department of Computational Arts at York University, where he founded the Alice Lab, and his research has been published in numerous journals and conferences.